

Fig. 4 | Co-resistance blocks define exploitable background

Phi correlations across E. coli resistance labels show that focal drug labels are embedded in structured resistance ecology.

	Cipro	Norflox	Amox-Clav	CRO	CAZ	FEP
Cipro	1.00	0.98	0.30	0.48	0.40	0.41
Norflox	0.98	1.00	0.34	0.42	0.37	0.48
Amox-Clav	0.30	0.34	1.00	0.34	0.38	0.33
CRO	0.48	0.42	0.34	1.00	0.88	0.80
CAZ	0.40	0.37	0.38	0.88	1.00	0.83
FEP	0.41	0.48	0.33	0.80	0.83	1.00

Strongest resistant blocks

Cipro / Norflox	phi=0.976	RR lift=3.99
CRO / CAZ	phi=0.884	RR lift=7.84
CAZ / FEP	phi=0.828	RR lift=10.21
CRO / FEP	phi=0.804	RR lift=9.39
Norflox / FEP	phi=0.479	RR lift=2.91

Interpretation

Models can exploit spectra associated with these resistant blocks without learning a focal drug in isolation.